② 3.3-1

$$\dot{x} = Ax + Bu \qquad A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ -1 & -3 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$sX = AX + BU$$

$$(s[I] - A)X = BU$$

$$A(s)X(s) = B(s)U(s)$$

$$X_i(s) = \frac{\det(A_i(s))}{\det(A(s))}$$

A_i is A with ith column replaced by $B(s)$

$$X_2(s) = \frac{\det\begin{bmatrix} s-1 & 0 & -1 \\ 0 & 1 & -2 \\ 1 & 1 & s+2 \end{bmatrix}}{\det\begin{bmatrix} s-1 & 0 & -1 \\ 0 & s & -2 \\ 1 & -3 & s+2 \end{bmatrix}} U(s)$$

$$\frac{X_2(s)}{U(s)} = \frac{(s-1)(s+2+2) + 0 + (-1)(0-1)}{(s-1)(s^2+2s+6) + 0 + (-1)(0-s)}$$

$$\frac{X_2(s)}{U(s)} = \frac{s^2+4s-s-4+1}{s^3+2s^2+6s-s^2-2s-6+s}$$

$$\boxed{\frac{X_2(s)}{U(s)} = \frac{s^2+3s-3}{s^3+s^2+5s-6}}$$

```
A = [ 1  0  1;
      0  0  2;
     -1 -3 -2];
B = [0;
     1;
     1];
C = [0 1 0];
D = 0;
sys = ss(A, B, C, D);
[num, den] = tfdata(tf(sys), 'v');
disp('Numerator Coefficients:');
disp(num);
disp('Denominator Coefficients:');
disp(den);
Numerator Coefficients:
         0    1.0000    3.0000   -3.0000

Denominator Coefficients:
    1.0000    1.0000    5.0000   -6.0000
```

verified ✓